```
# Import libraries
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

# Dataset load
df = pd.read_csv('student_performance_dataset.csv')  # apni file ka path

# Check data
print(df.head())
print(df.isnull().sum())

# Preprocessing
df = df.dropna()

# Suppose target = pass/fail (0/1)
X = df.drop('Pass_Fail', axis=1)
y = df['Pass_Fail']

# Encoding
X = pd.get_dummies(X)

# Train test split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, r

# Scaling

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Logistic Regression model

model = LogisticRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
from sklearn.metrics import accuracy_score, confusion_matrix, classificati

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Report:\n", classification_report(y_test, y_pred))
```

```
  Student_ID  Gender  Study_Hours_per_Week  Attendance_Rate  Past_Exam_Scores  \
0       S327    Male                    32        93.031601                78
1       S327    Male                    32        93.031601                78
2       S327    Male                    32        93.031601                78
3       S412  Female                    38        82.252708                95
4       S241  Female                    30        99.884631                98

  Parental_Education_Level Internet_Access_at_Home Extracurricular_Activities  \
0                Bachelors                     Yes                        Yes
1                Bachelors                     Yes                        Yes
2                Bachelors                     Yes                        Yes
3                Bachelors                      No                         No
4                  Masters                     Yes                         No

   Final_Exam_Score Pass_Fail
0                77      Pass
1                77      Pass
2                77      Pass
3                76      Pass
4                74      Pass
Student_ID                    0
Gender                        0
Study_Hours_per_Week          0
Attendance_Rate               0
Past_Exam_Scores              0
Parental_Education_Level      0
Internet_Access_at_Home       0
Extracurricular_Activities    0
Final_Exam_Score              0
Pass_Fail                     0
dtype: int64
Accuracy: 0.9436619718309859
Confusion Matrix:
 [[55  8]
 [ 0 79]]
Report:
              precision    recall  f1-score   support

        Fail       1.00      0.87      0.93        63
        Pass       0.91      1.00      0.95        79

    accuracy                           0.94       142
   macro avg       0.95      0.94      0.94       142
weighted avg       0.95      0.94      0.94       142
```